

Chapter 6

Transmittance Functions, Lenses, and Gratings

The beam sources implemented in Chapter 5 are for the most part simple apertures illuminated by a plane wave. They are modeled with real functions and, in effect, have a zero phase component. In this chapter functions are presented that create a more complicated field by altering the magnitude and/or phase of the field. The functions can be used to apply “tilt” or “focus” to a field, model the effect of a periodic structure, or model a lens. In general, these transmittance functions can be thought of as multiplying an incident field to create a desired effect; however, some represent well-known optical components such as a diffraction grating or a lens.

The functions discussed in this chapter provide considerable utility in their own right, but like the basic functions they can be combined to create more elaborate fields. As a matter of convenience these functions are described as part of the source, or as applied in the source plane. However, they can be applied in other planes; for example, the pupil of an imaging system, which is coming up in Chapter 7.

6.1 Tilt

An optical beam can be steered in a propagation simulation by applying a “tilt” to the beam wavefront. Suppose a tilt of angle of α is applied to a wavefront, as indicated in Fig. 6.1, where the dashed line represents the tilted wavefront of the beam and the arrow indicates the intended direction of propagation.

An expression for the dashed line in Fig. 6.1 is $z = -y \tan \alpha$. The intent is to convert this line to a phase front in the x - y plane at $z = 0$. This essentially requires replacing the position z with a phase quantity. As time progresses the wave moves in the positive z direction, but (as noted previously), the phase representation becomes more negative. This reverses the sign of the expression. The wavelength λ corresponds to 2π rad in the phase notation; so, using the wavenumber parameter $k = 2\pi/\lambda$, the phase function for producing the tilt is

$$\phi_y(x, y) = ky \tan \alpha . \quad (6.1)$$

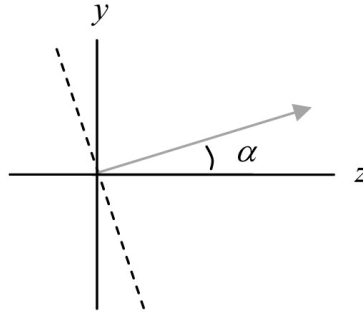


Figure 6.1 Wavefront tilt in the y - z plane.

More generally, to produce a tilt α relative to the z axis but in a radial direction defined by the angle θ in the x - y plane (see Fig. 6.2), one can use

$$\phi(x, y) = k(x \cos \theta + y \sin \theta) \tan \alpha, \quad (6.2)$$

where $\theta = \tan^{-1}(y/x)$. The transmittance function is, therefore,

$$t_A(x, y) = \exp[jk(x \cos \theta + y \sin \theta) \tan \alpha]. \quad (6.3)$$

Note that $\alpha = \tan^{-1}(r/z)$, where r is the radial distance in the observation plane from the origin to the beam aiming point (Fig. 6.2). The function `tilt` is listed here:

```

1  function[uout]=tilt(uin,L,lambda,alpha,theta)
2  % tilt phasefront
3  % uniform sampling assumed
4  % uin - input field
5  % L - side length
6  % lambda - wavelength
7  % alpha - tilt angle
8  % theta - rotation angle (x axis 0)

```

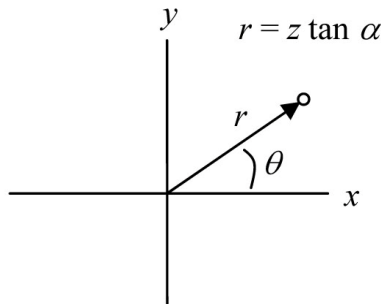


Figure 6.2 Beam-aiming point in the observation plane.

```

9  % uout - output field
10
11  [M,N]=size(uin);           %get input field array size
12  dx=L/M;                   %sample interval
13  k=2*pi/lambda;            %wavenumber
14
15  x=-L/2:dx:L/2-dx;         %coords
16  [X,Y]=meshgrid(x,x);
17
18  uout=uin.*exp(j*k*(X*cos(theta)+Y*sin(theta))...
19      *tan(alpha));           %apply tilt
20  end

```

In the expression for u_{out} (line 18), typing three periods and hitting enter allows the equation to be continued on the next line. Test the `tilt` function by returning to the “`sqr_beam`” example from Section 5.3. Save a new version (`sqr_beam_tilt`) and insert the following before the propagation call:

```

deg=pi/180;
alpha=5.0e-5; %rad
theta=45*deg;
[u1]=tilt(u1,L1,lambda,alpha,theta);

```

where Fig. 6.3 is the irradiance result after executing the example.

Sampling limitations also exist for this technique. As one might guess, if the tilt is large enough to translate the beam in the observation plane beyond the grid boundary, there will be trouble (see Exercise 6.1). To study this limitation, consider that tilt is a linear phase exponential applied to the source function U_1 . Assume a single-axis tilt, and apply the shift theorem to the transform of the source field to get

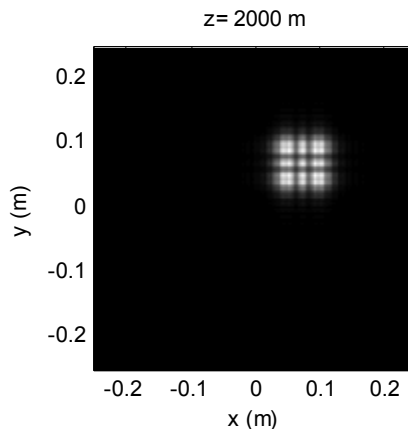


Figure 6.3 Irradiance pattern for the tilt example: $\alpha = 0.5 \mu\text{rad}$, $\theta = 45 \text{ deg}$.

$$\mathfrak{I}\{U_1(x_1, y_1) \exp(jkx_1 \tan \alpha)\} = G_1 \left(f_{x1} - \frac{\tan \alpha}{\lambda}, f_{y1} \right), \quad (6.4)$$

where $G_1 = \mathfrak{I}\{U_1\}$. If the source has a bandwidth of B_1 , then considering the spectrum is essentially shifted, the combined effective bandwidth for the field leaving the source is

$$B_1^{+T} = B_1 + \frac{\tan \alpha}{\lambda}. \quad (6.5)$$

This bandwidth can now be substituted for B_1 in the propagation criteria of Table 5.3. For example, if $\Delta x \leq \lambda z/L$, then the propagation criterion is $B_1^{+T} \leq 1/(2\Delta x)$. Using Eq. (6.5), and with some rearrangement, we get

$$\alpha \leq \lambda \left(\frac{1}{2\Delta x_1} - B_1 \right), \quad (6.6)$$

where α is assumed to be a small angle. Some comments about this criterion include the following:

- (a) The result is approximate as the specific interaction of the source and propagator phase is not accounted for in Eq. (6.5) or (6.6). For example, one may cancel some of the effects of the other.
- (b) If $\Delta x > \lambda z/L$ (“shorter distance”), a variety of artifacts can appear in the observation plane, such as “fringing” and asymmetries before the tilt angle α nears the bound.
- (c) If $\Delta x < \lambda z/L$ (“longer distance”), replace $1/2\Delta x_1$ with $L_1/(2\lambda z)$ in Eq. (6.6).

In the `sqr_beam_tilt` example, $\lambda = 0.5 \times 10^{-6}$ m and $\Delta x_1 = 2 \times 10^{-3}$ m. Using $B_1 \approx 98$ cycles/m, which was calculated in Eq. (5.10), the tilt angle criterion result is $\alpha \leq 7.6 \times 10^{-5}$ rad. Try this tilt angle and see what happens. Set `theta=0` and `alpha=7.6e-5`. The resulting beam should appear close to the array edge. Check the magnitude profile where some “wrap-around” effects are apparent with energy from the beam entering the opposite side of the grid (periodic extension!). To split the pattern across the grid boundary, let $\alpha = \lambda/2\Delta x_1 = 1.25 \times 10^{-4}$.

Try some shorter and longer propagation distances in `sqr_beam_tilt` and observe how the beam is affected as the tilt increases. In general, it is a good idea to work with tilt angles that are well within the limit set by Eq. (6.6). In typical simulations the maximum available tilt angle is quite small, which is consistent with the paraxial nature of the Fresnel propagator.

6.2 Focus

Another useful operation is converging (“focusing”) or diverging (“defocusing”) an optical beam. A beam with a spherical wavefront, as shown in Fig. 6.4, will converge to the position z_f on the z axis. We can proceed in the same manner as was done for tilt to find the converging phase front in the x - y plane at $z = 0$. This is given by

$$\phi_s(x, y) = -k\sqrt{z_f^2 + x^2 + y^2}. \quad (6.7)$$

As x or y increases the phase values become more negative, which indicates the wavefront off the axis *leads* (in time) the on-axis wavefront (see Section 4.3). Thus, the negative sign in Eq. (6.7) corresponds to a *converging* wavefront, as illustrated in Fig. 6.4. A positive sign corresponds to a *diverging* wavefront. Borrowing from the discussion of the Fresnel diffraction in Section 4.4.2, the application of the binomial approximation gives a parabolic phase front that approximates the spherical phase front:

$$\phi(x, y) = -\frac{k}{2z_f}(x^2 + y^2). \quad (6.8)$$

The transmittance function for focus is, therefore,

$$t_A(x, y) = \exp\left[-j\frac{k}{2z_f}(x^2 + y^2)\right]. \quad (6.9)$$

This is a phase chirp function of the same form that appears in Eq. (4.23) for the Fresnel impulse response function h , although the exponent sign is negative.

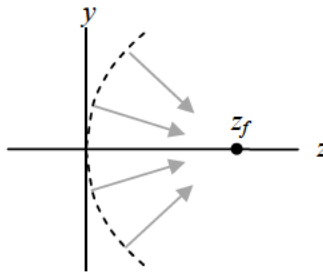


Figure 6.4 Converging wavefront.

A MATLAB function for applying focus follows:

```

1  function[uout]=focus(uin,L,lambda,zf)
2  % converging or diverging phase-front
3  % uniform sampling assumed
4  % uin - input field
5  % L - side length
6  % lambda - wavelength
7  % zf - focal distance (+ converge, - diverge)
8  % uout - output field
9
10 [M,N]=size(uin);           %get input field array size
11 dx=L/M;                    %sample interval
12 k=2*pi/lambda;              %wavenumber
13 %
14 x=-L/2:dx:L/2-dx;          %coords
15 [X,Y]=meshgrid(x,x);
16
17 uout=uin.*exp(-j*k/(2*zf)*(X.^2+Y.^2)); %apply focus
18 end

```

Try this on the `sqr_beam` example (new file `sqr_beam_focus`). Insert the following before the propagation call and run the script:

```

zf=2000;
[u1]=focus(u1,L1,lambda,zf);

```

The result is shown in Fig. 6.5. That looks like a pretty good focus! In this example the focal distance is the same as the propagation distance, so a small spot is expected. The pattern is, in fact, a scaled Fraunhofer pattern. (Check out Section 6.3.) Try some other focal distances—see what happens. Can you get the pattern to expand to fill the observation plane grid? A negative focus value puts the focal point in a virtual position behind the plane and causes a diverging wave (Fig. 6.6). Try it!

Multiplying a source field by Eq. (6.9) has the effect of increasing the source bandwidth. The increase in bandwidth is roughly the source support radius $D_1/2$ over $\lambda|z_f|$, so the combination effective bandwidth is approximately

$$B_1^{+F} \approx B_1 + \frac{D_1/2}{\lambda|z_f|}. \quad (6.10)$$

For $B_1^{+F} \leq 1/(2\Delta x)$, then with some rearranging, a bound is obtained for the focal distance:

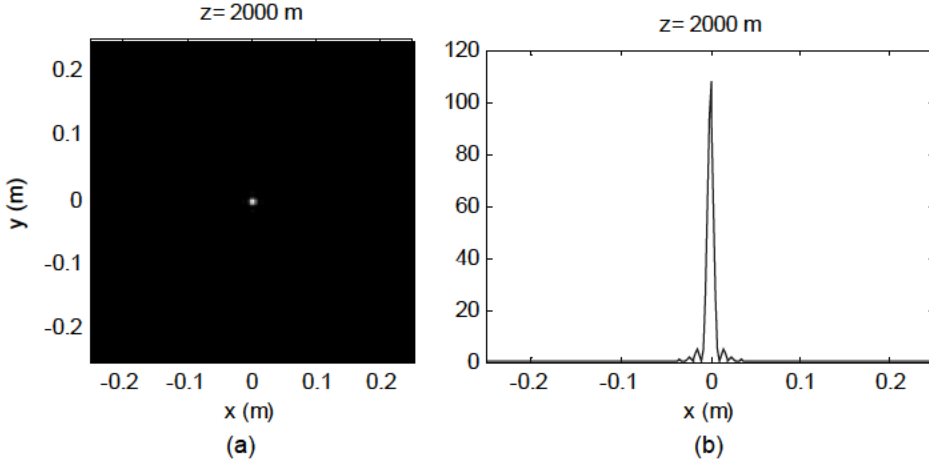


Figure 6.5 (a) Image irradiance and (b) profile for the focus example.

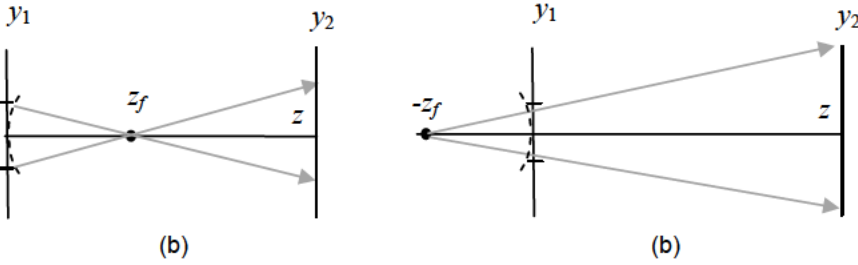


Figure 6.6 Geometrical ray diagram for (a) a converging wavefront and (b) a diverging wavefront at the source plane.

$$|z_f| \geq \frac{D_1/2}{\lambda} \left(\frac{1}{2\Delta x_1} - B_1 \right)^{-1}. \quad (6.11)$$

As with the tilt angle criterion, Eq. (6.11) is approximate because the exact interaction of the source and propagator phase is not considered.

For the `sqr_beam_focus` example, $\lambda = 0.5 \times 10^{-6}$ m, $\Delta x_1 = 2 \times 10^{-3}$ m, $B_1 \approx 98$ cycles/m and $D_1 = 0.102$ m, which leads to $|z_f| \geq 671$ m. Test this condition with the `sqr_beam_focus` code. When $z_f = -671$ m, the beam is diverging and clearly is pushing over the array boundaries in the observation plane. When $z_f = +671$ m, the pattern is smaller since the beam went through a focus before the observation plane (Fig. 6.6). The pattern looks reasonable; however, phase aliasing errors are just starting to creep in on the array edges. This can be seen in the unwrapped phase profile of the observation plane field.

6.3 Lens

A lens is an optical element that uses refraction to focus or diverge light. The transmittance function for an ideal, simple lens is given by¹

$$t_A(x, y) = P(x, y) \exp \left[-j \frac{k}{2f} (x^2 + y^2) \right], \quad (6.12)$$

where f is known as the focal length and $P(x, y)$ is the pupil function. This is essentially the same complex exponential defined for focus with z_f replaced by f . A positive focal length produces a converging wavefront from a plane-wave input and a negative focal length produces a diverging wavefront. The pupil function accounts for the physical size of the lens—the opening available to collect light. For example, the most common lens pupil function is a circle

$$P(x, y) = \text{circ} \left(\frac{\sqrt{x^2 + y^2}}{w_L} \right), \quad (6.13)$$

where w_L is the radius of the lens aperture (not to be confused with the radius of curvature of the wavefront that exits the lens).

It is not always practical to implement the transmittance function of Eq. (6.12) in Fresnel propagation as was done for the focus example. This is because the focal length f is governed by the same criterion as z_f given in Eq. (6.11), and since f tends to be relatively short, a large number of samples are required. Assume a plane wave incident on the lens, which implies $B_1 \rightarrow 0$, then with some algebra the expression in Eq. (6.11) leads to

$$\frac{|f|}{D_L} \geq \frac{\Delta x}{\lambda}, \quad (6.14)$$

where $D_L = 2w_L$. The ratio $|f|/D_L$ is known as the focal ratio, or the *f-number*, indicated by $f/\#$. With this substitution,

$$f/\# \geq \frac{\Delta x}{\lambda}. \quad (6.15)$$

Practical lenses have $f/\#$ s ranging from perhaps 2 to roughly 50 and diameters $2w_L$ of a few millimeters to maybe 100 mm. Take a typical value, $f/\# = 10$ (usually written $f/10$) and a diameter of 25 mm. Assume visible light $\lambda = 0.5 \times 10^{-6}$ m, then Eq. (6.15) yields

$$\Delta x \leq 5 \times 10^{-6} \text{ m.} \quad (6.16)$$

Furthermore, to implement Fresnel propagation, the array side length L needs to at least span the lens diameter. Thus, the linear number of samples in an array required to model this lens is

$$M \geq \frac{L}{\Delta x} = \frac{25 \times 10^{-3}}{5 \times 10^{-6}} = 5000, \quad (6.17)$$

which is a rather large array. Thus, modeling lenses directly with the Fresnel propagator is usually practical only for large $f/\#$ s.

However, all is not lost for smaller $f/\#$ s. If the field incident on the lens is $U_1(x_1, y_1)$, then the field exiting the lens is $U_1(x_1, y_1)t_A(x_1, y_1)$. Insert this into Eq. (4.25) for $U_1(x_1, y_1)$ and set $z = f$. The chirp functions in the integral cancel, and the result is

$$U_2(x_2, y_2) = \frac{\exp(jkf)}{j\lambda f} \exp\left[j \frac{k}{2f}(x_2^2 + y_2^2)\right] \times \iint U_1(x_1, y_1) P(x_1, y_1) \exp\left[-j \frac{2\pi}{\lambda f}(x_2 x_1 + y_2 y_1)\right] dx_1 dy_1. \quad (6.18)$$

The expression in Eq. (6.18) shows the field at the *focal plane* of an ideal positive lens is simply the *Fraunhofer pattern* of the incident field with $z = f$.

Therefore, to find the field or irradiance pattern in the focal plane of a positive lens, including one with a small $f/\#$, the function “prop_FF” from the previous chapter can be applied replacing z with f .

Take the parameters from the $f/10$ lens example and assume U_1 is a unit amplitude plane wave. Select $M = 250$ and $L = 250$ mm. The irradiance pattern in Fig. 6.7 is generated for the focal plane using prop_FF (see Exercise 6.4). Try it! The focused irradiance pattern formed with an ideal circular-shaped lens, such as shown in Fig. 6.7, is known as the Airy pattern.

A special case of interest is when the source field is located in the front focal plane of a positive lens, a distance f from the lens (Fig. 6.8). For this arrangement the field at the focal plane is¹

$$U_2(x_2, y_2) = \frac{\exp(jkf)}{j\lambda f} \times \iint U_1(x_1, y_1) P(x_1 + x_2, y_1 + y_2) \exp\left[-j \frac{2\pi}{\lambda f}(x_2 x_1 + y_2 y_1)\right] dx_1 dy_1. \quad (6.19)$$

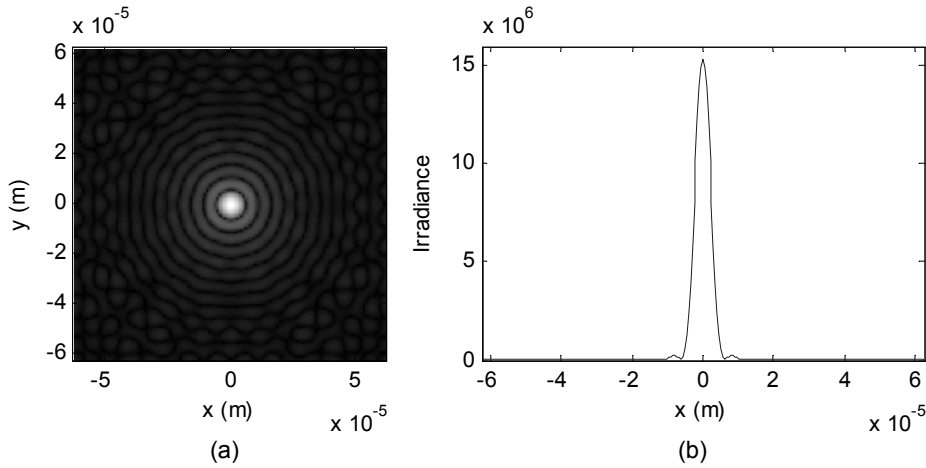


Figure 6.7 (a) Image irradiance and (b) x-axis profile for the lens Fraunhofer pattern. The fourth root is applied for (a). The large peak irradiance value in (b) is because all of the power in the unit amplitude field incident on the lens is being focused to a very small area.

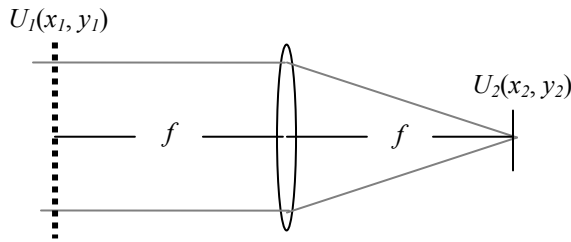


Figure 6.8 “Fourier transform” lens arrangement.

The chirp phase factor out front is now gone, so the focal plane field is a scaled *Fourier transform* of the input field. The arguments in the pupil function account for *vignetting*, which is a loss of light for off-axis points in the input field due to the finite pupil size. The effect of vignetting is reduced if the lens pupil is oversized compared to the support of the input field.

6.4 Gratings and Periodic Functions

A grating is an optical component that has a spatially periodic structure. Incident light diffracts either in transmission or reflection from the structure, and the colors (wavelength components) of the light become spatially separated some distance from the grating. Gratings are commonly used in spectrometers for examining the wavelength spectrum of an optical signal or in spectrophotometers that measure the spectral characteristics of an optical component. The diffraction pattern from a grating is usually observed in the Fraunhofer region.

6.4.1 Cosine magnitude example

A conventional grating has grooves cut into its surface that impart a magnitude and/or phase disturbance to the incident wave. To demonstrate modeling of periodic functions like those in gratings, start with an amplitude transmittance function given by

$$t_A(x, y) = \frac{1}{2} \left[1 - \cos \left(2\pi \frac{x}{P} \right) \right] \text{rect} \left(\frac{x}{D_1} \right) \text{rect} \left(\frac{y}{D_1} \right). \quad (6.20)$$

Figure 6.9 illustrates a 1D profile of this grating. In Eq. (6.20) the grating is defined within the 2D area $D_1 \times D_1$. The cosine pattern is only a function of x and has a period P . Typically, $P \ll D_1$. When illuminated by a unit amplitude plane wave, the source field is $U_1(x_1, y_1) = t_A(x_1, y_1)$. The Fraunhofer pattern is created using a lens (or mirror) of focal length f . The grating is simulated in the script “grating_cos,” where the first part of this code is shown here:

```

1  % grating_cos diffraction grating example
2
3  lambda=0.5e-6;    %wavelength
4  f=0.5;            %focal distance
5  P=1e-4;           %grating period
6  D1=1.02e-3;       %grating side length
7
8  L1=1e-2;          %array side length
9  M=500;            %# samples
10 dx1=L1/M;
11 x1=-L1/2:dx1:L1/2-dx1; %source coords
12 [X1,Y1]=meshgrid(x1,x1);
13
14 % Grating field and irradiance
15 u1=1/2*(1-cos(2*pi*X1/P)).*rect(X1/D1).*rect(Y1/D1);
16
17 % Fraunhofer pattern
18 [u2,L2]=propFF(u1,L1,lambda,f);
19 dx2=L2/M;
20 x2=-L2/2:dx2:L2/2-dx2; y2=x2;%obs coords
21 I2=abs(u2).^2;
```

It is critical that the periodic function be sampled adequately. The number of samples that span the periodic function (the length P) is

$$\# \text{samples across } P = M \frac{P}{L_1}. \quad (6.21)$$

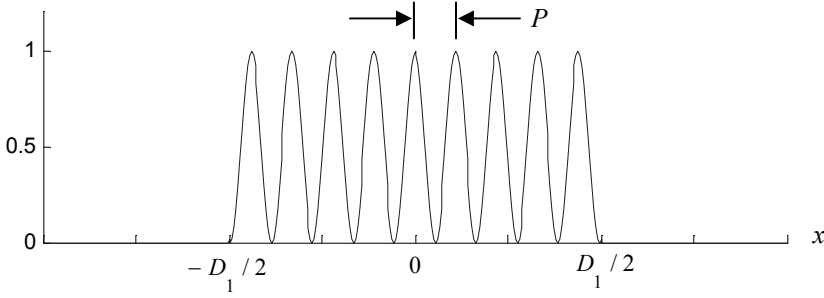


Figure 6.9 Cosine grating profile.

In the case of `grating_cos`, $MP/L_1 = 5$ indicates that five samples span each cosine cycle, which is okay. At least two are required to satisfy the sampling theorem. The values for L_1 , D_1 , P , and M were also selected in `grating_cos` to provide a clear display of the Fraunhofer pattern. Furthermore, the value for D_1 provides an odd number of samples ($D_1/dx_1 = 51$) across the `rect` functions, which is consistent with our `rect` function definition. The lens focal length was arbitrarily chosen as $f = 0.5$ m. Figure 6.10 shows irradiance images of the source plane ($\mathbb{I}_1$) and the observation plane ($\mathbb{I}_2$) and an x -axis profile in the observation plane. The central feature is known as the zero order and the two side features are the -1 and $+1$ “first-order” peaks.

To make sure the simulation is working properly, the results can be compared with the analytic expression for the Fraunhofer pattern. First, the Fourier transform of the source field is required:

$$\begin{aligned} \mathfrak{F}\{U_1(x_1, y_1)\} &= \frac{1}{2} \left[\delta(f_{x1}, f_{y1}) - \frac{1}{2} \delta\left(f_{x1} + \frac{1}{P}, f_{y1}\right) - \frac{1}{2} \delta\left(f_{x1} - \frac{1}{P}, f_{y1}\right) \right] \\ &\quad \otimes D_1^2 \text{sinc}(D_1 f_{x1}) \text{sinc}(D_1 f_{y1}). \end{aligned} \quad (6.22)$$

Then perform the convolution:

$$\begin{aligned} \mathfrak{F}\{U_1(x_1, y_1)\} &= \frac{D_1^2}{2} \text{sinc}(D_1 f_{y1}) \\ &\quad \times \left\{ \text{sinc}(D_1 f_{x1}) - \frac{1}{2} \text{sinc}\left[D_1 \left(f_{x1} + \frac{1}{P}\right)\right] - \frac{1}{2} \text{sinc}\left[D_1 \left(f_{x1} - \frac{1}{P}\right)\right] \right\}. \end{aligned} \quad (6.23)$$

Substitute $x_2/\lambda f \rightarrow f_{x1}$ and $y_2/\lambda f \rightarrow f_{y1}$ and apply the multipliers to get the Fraunhofer field:

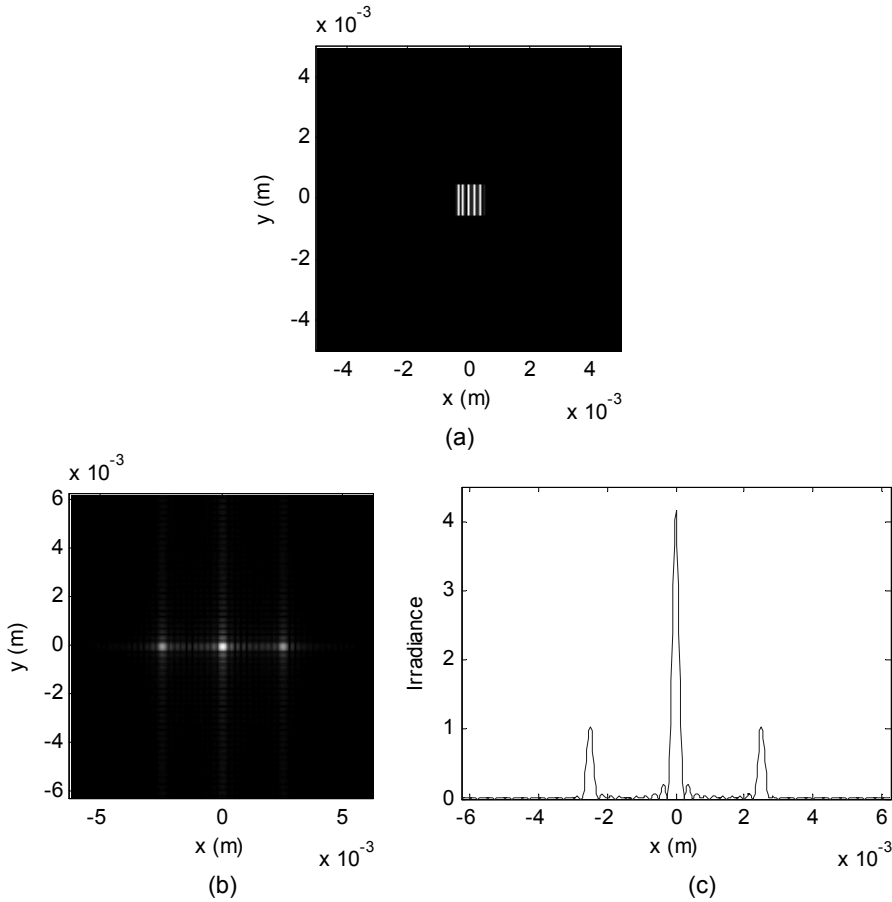


Figure 6.10 Results for `grating_cos`: (a) source irradiance; (b) Fraunhofer irradiance (contrast enhanced with third root); and (c) x-axis profile of Fraunhofer irradiance. [The display in (a) does not resolve the grating periodic features.]

$$\begin{aligned}
 U_2(x_2, y_2) = & \frac{\exp(jkf)}{j\lambda f} \exp\left[j\frac{k}{2f}(x_2^2 + y_2^2)\right] \frac{D_1^2}{2} \operatorname{sinc}\left(\frac{D_1}{\lambda f} y_2\right) \\
 & \cdot \left\{ \operatorname{sinc}\left(\frac{D_1}{\lambda f} x_2\right) - \frac{1}{2} \operatorname{sinc}\left[\frac{D_1}{\lambda f} \left(x_2 + \frac{\lambda f}{P}\right)\right] - \frac{1}{2} \operatorname{sinc}\left[\frac{D_1}{\lambda f} \left(x_2 - \frac{\lambda f}{P}\right)\right] \right\}.
 \end{aligned} \tag{6.24}$$

The irradiance is the squared magnitude of Eq. (6.24). The following script portion evaluates the analytic irradiance result:

```

%analytic
[X2,Y2]=meshgrid(x2,y2);
lf=lambda*f;
u2a=(1/lf)*D1^2/2*sinc(D1/lf*Y2)...
   .*(sinc(D1/lf*X2)-1/2*sinc(D1/lf*(X2+lf/P))...

```

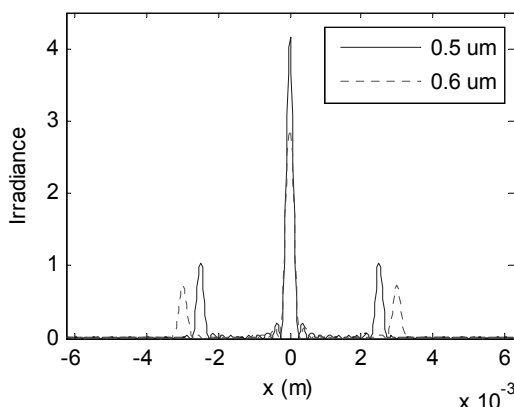


Figure 6.11 Irradiance profiles for `grating_cos` for $\lambda = 0.5 \mu\text{m}$ and $0.6 \mu\text{m}$.

```
-1/2*sinc(D1/lf*(X2-lf/P));  
I2a=abs(u2a).^2;
```

The front complex exponential terms for the Fraunhofer pattern were not coded up since only the irradiance is being examined—but don’t forget $1/\lambda f$. Try this and see if the discrete and analytic plots come out the same.

The main application for a grating is wavelength separation. Figure 6.11 shows superimposed curves for $\lambda = 0.5 \mu\text{m}$ and $\lambda = 0.6 \mu\text{m}$. The first-order peaks are clearly separated. To produce these curves, run the code for one wavelength and in the command window store `I2` and `x2` in temporary arrays (for example, `I2p5=I2; x2p5=x2;`). Rerun the code for the other wavelength and execute the plot function for both curves under the Command Window using `plot(x2p5,I2p5(M/2+1,:),x2,I2(M/2+1,:))`. The `x2` scale is different for each wavelength—don’t use the same `x2` vectors for the two curves! “Illuminating” a larger grating area narrows the orders and results in better spectral resolution. You can test this by making D_1 larger.

6.4.2 Square-wave magnitude example

Another example of a grating is illustrated in Fig. 6.12. The transmittance function is given by

$$t_A(x, y) = \left[\text{rect}\left(\frac{x}{P/2}\right) * \frac{1}{P} \text{comb}\left(\frac{x}{P}\right) \right] \text{rect}\left(\frac{x}{D_1}\right) \text{rect}\left(\frac{y}{D_1}\right), \quad (6.25)$$

where P is the period of a square-wave amplitude pattern that stretches in the x direction and lies within the 2D area of $D_1 \times D_1$. This grating is simulated in the following script “`grating_sqr`” (only a portion is shown):

```

1  % grating_sqr diffraction grating example
2
3  lambda=0.5e-6; %wavelength
4  f=0.5;         %focal distance
5  P=1e-4;        %grating period
6  D1=1e-3;       %grating side length
7
8  L1=1e-2;       %array side length
9  M=1000;        %# samples
10 dx1=L1/M;
11 x1=-L1/2:dx1:L1/2-dx1; %source coords
12 [X1,Y1]=meshgrid(x1,x1);
13
14 % construct grating field
15 fc=fft(fftshift(ucomb(x1/P)));
16 fr=fft(fftshift(rect(x1/(P/2))));
17 ux=ifftshift(ifft(fc.*fr)); %1D conv rect & comb
18 u1=repmat(ux,M,1);          %replicate to 2D
19 u1=u1.*rect(X1/D1).*rect(Y1/D1); %set size

```

... (then apply the Fraunhofer propagator).

In this example the `ucomb` function is used to create a 1D periodic sequence of unit sample delta functions (defined in Appendix C). This delta is defined as a unit value at the coordinate of interest. The `ucomb` function truncates the input values at the sixth decimal position, so small round-off error will not cause problems in placing the unit samples.

For the `ucomb` function to work properly, the vector coordinates must be such that a sample is found at each position where the delta function is needed. Thus, $P/dx1$ needs to be an integer. In this case, $dx1=1 \times 10^{-5}$ m. Therefore the unit values appear in the vector `fc` for every 10 samples ($P/dx1 = 10$). An odd number of samples is arranged across each of the periodic rect functions ($P/2 = 5$) to be consistent with the `rect` function definition.

The approach for `grating_sqr` is to perform a 1D convolution (`fft` rather

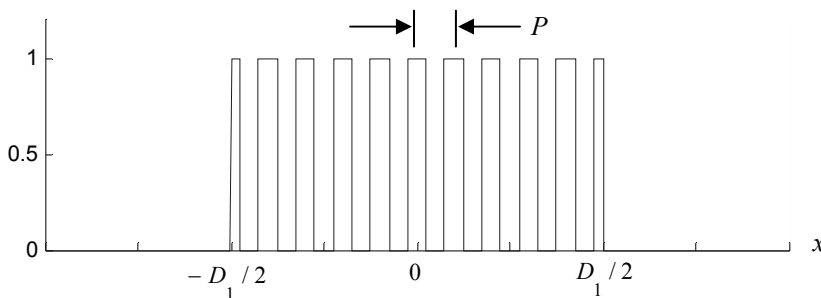


Figure 6.12 Square-wave grating profile.

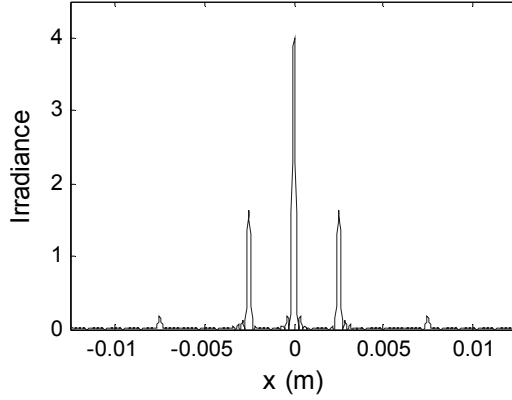


Figure 6.13 Fraunhofer irradiance profile for the grating modeled in `grating_sqr`.

than `fft2`) with the `ucomb` and `rect` sequences. A scaling multiplier is not needed on the convolution result because (a) the `ifft` correctly scales for one `fft` (for example, `fr`) and (b) each unit sample delta behaves as though it has an area of $1/dx \times 1$, which cancels the $dx \times 1$ that arises from the second `fft`. The `repmat` function is a quick way to fill the rows of the 2D array `u1` with the vector `ux`.

The field created by the grating is propagated using `propFF`, and the profile for `I2` is shown in Fig. 6.13. It is similar to the cosine grating but with some additional higher-order peaks. Again, analytic theory can be used to verify the simulation. To find the Fraunhofer pattern, take the Fourier transform of the field and simplify as follows:

$$\begin{aligned}
 \mathfrak{F}\{U_1(x_1, y_1)\} &= \left[\frac{P}{2} \text{sinc}\left(\frac{P}{2} f_{x1}\right) \text{comb}(P f_{x1}) \delta(f_{y1}) \right] \\
 &\quad \otimes [D_1 \text{sinc}(D_1 f_{x1}) D_1 \text{sinc}(D_1 f_{y1})] \\
 &= \left[\frac{P}{2} \text{sinc}\left(\frac{P}{2} f_{x1}\right) \frac{1}{P} \sum_{n=-\infty}^{\infty} \delta\left(f_{x1} - \frac{n}{P}\right) \delta(f_{y1}) \right] \\
 &\quad \otimes [D_1 \text{sinc}(D_1 f_{x1}) D_1 \text{sinc}(D_1 f_{y1})] \\
 &= \frac{D_1^2}{2} \text{sinc}(D_1 f_{y1}) \sum_{n=-\infty}^{\infty} \text{sinc}\left(\frac{n}{2}\right) \text{sinc}\left[D_1 \left(f_{x1} - \frac{n}{P}\right)\right]. \quad (6.26)
 \end{aligned}$$

The Fraunhofer field is given by

$$U_2(x_2, y_2) = \frac{\exp(jkf)}{j\lambda f} \exp\left[j\frac{k}{2f}(x_2^2 + y_2^2)\right] \frac{D_1^2}{2} \operatorname{sinc}\left(\frac{D_1}{\lambda f} y_2\right) \cdot \sum_{n=-\infty}^{\infty} \operatorname{sinc}\left(\frac{n}{2}\right) \operatorname{sinc}\left[\frac{D_1}{\lambda f}\left(x_2 - n\frac{\lambda f}{P}\right)\right]. \quad (6.27)$$

Below is part of the code to generate a 1D irradiance slice for Eq. (6.27). A `for` loop is used for the summation and 11 terms are computed ($n = -5$ to 5). More terms do not make a noticeable difference in the result:

```
lf=lambda*f;
u2a=zeros(1,M);
for n=-5:5
    ut=sinc(n/2)*sinc(D1/lf*(x2-n*lf/P));
    u2a=u2a+ut;
end
u2a=D1^2/(2*lf)*u2a;
```

Implement this code at the end of `sqr_grating`. Compare the results with the numerical simulation. Can you get them to match?

6.4.3 One-dimensional model

The gratings examined thus far have no y directional periodic dependence. When this is the case, a 1D model (where $y = 0$) can be used to analyze the most critical part of the grating response, which is along the x axis. To do this, the y -dependent terms are removed, `meshgrid` is no longer required, and the function `repmat` is not required. Also, a 1D Fraunhofer calculation is required. The advantage of 1D modeling is that larger vectors can be used; so, more overall width and cycles across the grating can be modeled.

Return to the `grating_sqr` example and make the changes listed above for a 1D result. Call this new script “`grating_sqr1D`.” Let $M=2200$, but leave all of the physical parameter definitions the same as before. Be sure to remove `rect(Y1/D1)` from line 19. The 2D function `propFF` is removed and the 1D Fraunhofer pattern can be calculated using

```
lf=lambda*f;
u2=sqrt(1/lf)*ifftshift(fft(fftshift(u1)))*dx1;
L2=lf/dx1;
```

Figure 6.14 shows profiles of the 1D Fraunhofer pattern. The greater number of samples results in an increase of the side length of the Fraunhofer pattern. The expanded view in Fig. 6.11(b) shows the relative size and position of the orders are the same as in the original result in Fig. 6.10. The overall magnitudes are

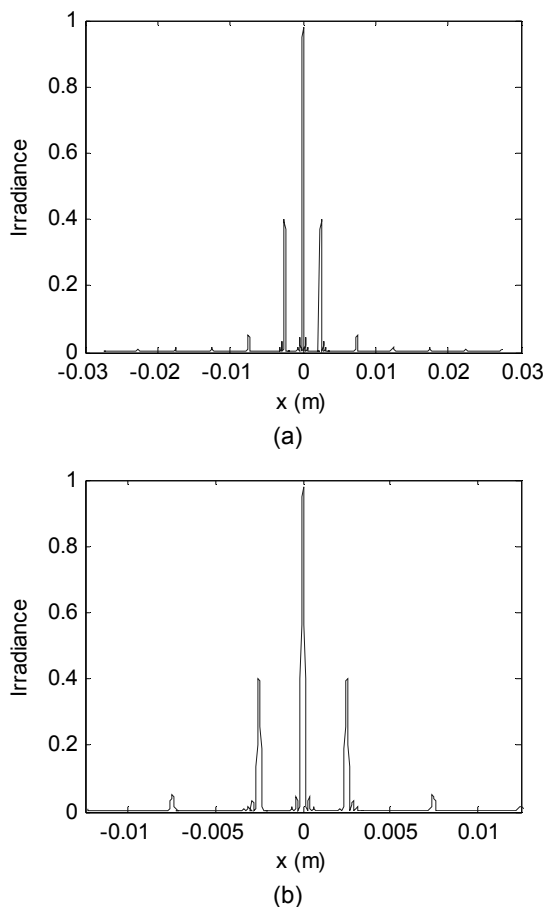


Figure 6.14 1D Fraunhofer irradiance profile for the grating modeled in `grating_sqr`: (a) full vector view; (b) expanded view.

different because the 1D case does not account for power associated with the second dimension. More samples across the periodic features in the 1D model produce fewer artifacts (small ripples) in the result compared to the 2D result.

6.4.4 Periodic model

How about modeling a grating that is essentially infinite in extent? Although not a practical device, it is not uncommon for a beam of light incident on a grating to cover hundreds or thousands of the periodic cycles—more than anything modeled here so far. But, more importantly, for an infinite-support grating the diffracted orders for an infinite (analytic) grating are delta functions and the multipliers for these delta functions indicate the relative amount of optical power that is directed to each order. For optical spectral analysis, high diffraction efficiency into the first order is usually desired.

Suppose the grating structure fills the array ($D_1 = L_1$) and the periodic features are carefully arranged such that an *integer number of periods* P exactly spans the array, or

$$\frac{L_1}{P} = \text{integer} . \quad (6.28)$$

This case is illustrated in Fig. 6.15. The discrete Fourier transform, applied to obtain the Fraunhofer pattern, produces a result that is associated with repeating copies of the input array (periodic extension). Since the source is arranged to be perfectly continuous at the array boundaries, the result is the transform of an infinite periodic structure.

Return to Section 6.4.1 and the `grating_cos` example. Set $D_1 = 1\text{e-}2$, which is the value of the side length L_1 . Since $L_1/P = 100$, there are exactly 100 cycles of the cosine function across the array. Executing the script yields sample delta functions at the diffractive order positions. To find the fraction of optical power in each order, compute the optical power at each sample in the observation

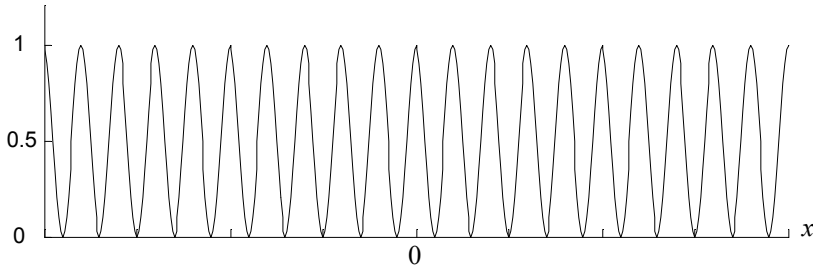


Figure 6.15 Cosine grating profile with an integer number of periods occupying the vector.

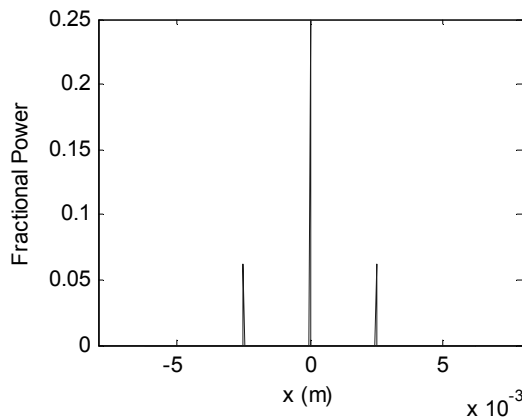


Figure 6.16 Fractional power result for a periodic model of the `grating_cos` example.

plane (irradiance integrated over the sample interval area) and divide by the total power transmitted by the grating (source field magnitude squared, integrated over the grating area). For `grating_cos` this operation looks like

$$F_{pwr} = I_2 * dx_2 * dx_2 / (L_1^2) ;$$

where a unit source field magnitude is assumed. A display of F_{pwr} is shown in Fig. 6.16 for the `grating_cos` example; 25% of the power is in the zero order, and 6.25% is found in each of the $\pm$ first orders.

6.5 Exercises

- 6.1** The diagram in Fig. 6.17 illustrates the propagation of a field at an angle α relative to the optical axis. The field support D_2 just reaches to the edge of observation plane side length, a distance $L/2$ from the axis. Use this arrangement to derive the same tilt criterion as defined in Eq. (6.6). Assume critical sampling and Fraunhofer propagation. For Fraunhofer propagation, $D_2 = 2B_1\lambda z$.
- 6.2** Demonstrate both tilt and focus simultaneously for the `sqr_beam` example. Use the TF approach with $z = 2000$ m. Assume $\alpha = 5.0 \times 10^{-5}$ rad, $\theta = 45$ deg for tilt and $z_f = 4000$ m for focus.
- 6.3** Among other applications, cylindrical lenses are commonly used to change an oblong-shaped laser diode beam into a more circular beam.
- (a) Create a function to produce a *cylindrical focus* (focus in only one transverse axis).
 - (b) Demonstrate the cylindrical focus with the `sqr_beam` example, $z_f = 2000$ m.

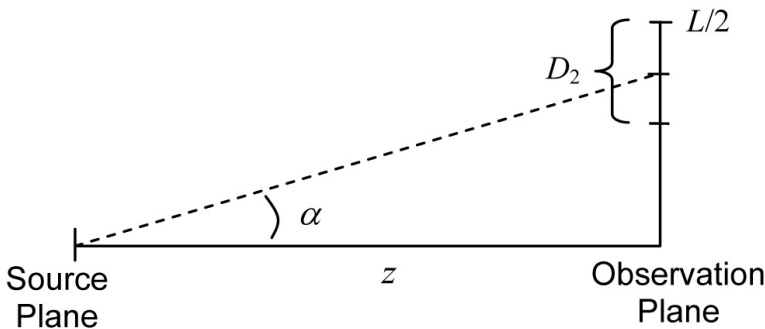


Figure 6.17 Diagram for propagation at an angle α .

- (c) Is it generally possible to describe the irradiance in the focal plane of a cylindrical lens in terms of a Fraunhofer pattern as was done in Eq. (6.18) or (6.19) for a spherical lens?

6.4 Airy pattern:

- (a) Generate the Airy pattern shown in Fig. 6.7 for an ideal $f/10$ lens ($f = 250$ mm and diameter $2w_L = 25$ mm). Use $M = 250$, $L = 250$ mm, and $\lambda = 0.5$ μm . Derive the analytic result and compare with the simulation result in a profile plot.
- (b) Demonstrate the effect in the focal plane of reducing the diameter of the lens pupil.

6.5 Zone plate: Zone plates can be used in parts of the electromagnetic spectrum where glass or other materials are not transparent; but, for what purpose? Let's find out. Consider the transmittance function

$$t_A(r) = \frac{1}{2} \left[1 + \cos\left(\frac{k}{2f} r^2\right) \right] \text{circ}\left(\frac{r}{w}\right), \quad (6.29)$$

where $r = \sqrt{x^2 + y^2}$. The transmittance of this plate is illustrated in Fig. 6.18.

- (a) Rewrite Eq. (6.29) using complex phasor notation. If a plane wave illuminates this plate, how do you expect the transmitted field to behave?
- (b) Derive a sampling criterion for rendering this plate in a discrete array (check out Appendix A). Create a Fresnel propagation simulation for this plate in the source plane. Assume the following: unit amplitude

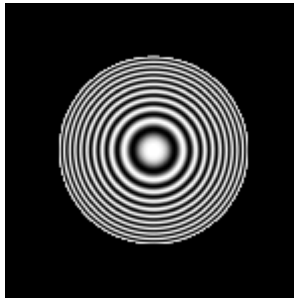


Figure 6.18 Zone plate transmittance illustration.

plane-wave illumination, $L_1 = 50$ mm, $M = 500$, $\lambda = 0.5$ μm , $w = 6.25$ mm, and $f = 10$ m. The choice of an extremely large value of f relative to the plate radius is necessary for sampling, and also to provide a magnified pattern at the observation plane.

- (c) Show whether the sampling criterion for the plate is satisfied.
- (d) Examine the propagation sampling for $z = 10$ m. Should the transfer function or impulse response approach be used?
- (e) Simulate the irradiance patterns for propagation distances of 5, 8, 10, and 12 m. Display the patterns and profiles.

Another type of zone plate is given by

$$t_A(r) = \frac{1}{2} \left\{ 1 + \text{sign} \left[\cos \left(\frac{k}{2f} r^2 \right) \right] \right\} \text{circ} \left(\frac{r}{w} \right), \quad (6.30)$$

where the sign function is

$$\text{sign}(x) = \begin{cases} -1, & x < 0. \\ 0, & x = 0. \\ 1, & x > 0. \end{cases} \quad (6.31)$$

This plate has a binary transmittance (either 1 or 0) and can be made with rings of suspended opaque material.

- (f) Simulate this plate and repeat part (e) using the same parameters as for the first plate (...you might check to see if MATLAB has a “sign” function...).
- (g) Compare the irradiance profiles for the two plates at the focal plane ($z = 10$ m). Which is more efficient?
- (h) Compare the zone plate profiles with the focal plane irradiance profile for a positive lens of the same characteristics.

6.6 How critical is the sampling arrangement for periodic functions? Do some testing! Change the number of samples in the `grating_sqr` example, for example, by +2. What happens? Try some other values (make sure M is still even so that other numerical issues are not also happening). What is the next value above $M = 1000$, where the periodic functions are again sampled appropriately?

6.7 Phase gratings: Unlike an amplitude grating, an ideal phase grating does not attenuate the incident light. Diffraction occurs because of periodic optical path length changes across the grating. A reflection grating can be modeled in the computer as a phase grating. Alter the `grating_cos` or

`grating_sqr` code to compute the Fraunhofer pattern for the transmittance functions that follow. Adjust the factor m ; for example, π , $\pi/2$, and $\pi/4$, and notice the effect on the Fraunhofer pattern.

$$(a) \ t_A(x, y) = \exp \left[jm \cos \left(2\pi \frac{x}{P} \right) \right] \text{rect} \left(\frac{x}{D_1} \right) \text{rect} \left(\frac{y}{D_1} \right),$$

$$(b) \ t_A(x, y) = \exp \left[jm \text{rect} \left(\frac{x}{P/2} \right) * \frac{1}{P} \text{comb} \left(\frac{x}{P} \right) \right] \text{rect} \left(\frac{x}{D_1} \right) \text{rect} \left(\frac{y}{D_1} \right),$$

$$(c) \ t_A(x, y) = \exp \left[jm \Lambda \left(\frac{x}{P/2} \right) * \frac{1}{P} \text{comb} \left(\frac{x}{P} \right) \right] \text{rect} \left(\frac{x}{D_1} \right) \text{rect} \left(\frac{y}{D_1} \right).$$

6.8 Consider the following grating transmittance function:

$$t_A(x, y) = \left[\Lambda \left(\frac{x}{P/2} \right) * \frac{1}{P} \text{comb} \left(\frac{x}{P} \right) \right] \text{rect} \left(\frac{x}{D_1} \right). \quad (6.32)$$

- (a) Implement this grating in a 1D simulation and compute the Fraunhofer pattern. Assume a grating frequency of 500 lines/mm [$P = (1/500) \times 10^{-3}$ m], $D_1 = 0.1$ mm, $\lambda = 0.5$ μ m, and $f = 0.5$ m. Choose the vector size and other sampling parameters. [Hint: An even number of samples—for example, 8—across each triangle function works well in this case.]
- (b) Derive a theoretical result for the Fraunhofer pattern for this grating (in 1D). Compare this result with the numerical result of part (a).
- (c) Implement the grating in an infinite periodic model. Plot the Fraunhofer pattern profile. Plot the power percentage result.

6.6 References

1. J. W. Goodman, *Introduction to Fourier Optics*, 3rd Ed., Roberts & Company, Greenwood Village, CO (2005).